

# Pulmonary Alveolar Proteinosis With Hypogammaglobulinemia (PAPHG): A Comprehensive Disease Characteristics Report

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**Disease:** Pulmonary Alveolar Proteinosis With Hypogammaglobulinemia (PAPHG) **MONDO ID:** MONDO:0020840 · **OMIM:** #618042 · **Category:** Mendelian (autosomal dominant) **Causal gene:** *OAS1* (2'-5'-oligoadenylate synthetase 1), 12q24.13, NCBI Gene 4938, HGNC:8086 **Also known as:** Immunodeficiency 100 with pulmonary alveolar proteinosis and hypogammaglobulinemia (IMD100)

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## Summary

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Pulmonary Alveolar Proteinosis with Hypogammaglobulinemia (PAPHG; OMIM #618042, MONDO:0020840) is an **ultra-rare, autosomal-dominant Mendelian disorder** caused by **heterozygous gain-of-function (GoF) missense variants in *OAS1***, the gene encoding 2'-5'-oligoadenylate synthetase 1 on chromosome 12q24.13. The disease was first defined genetically in 2018 when whole-exome sequencing identified a heterozygous *OAS1* missense variant segregating in three affected siblings, plus two de novo variants in unrelated infants, after causative variants in all known PAP genes (*SFTPB*, *SFTPC*, *ABCA3*, *CSF2RA*, *CSF2RB*, *GATA2*) had been excluded (Cho et al. 2018, [PMID: 29455859](#)). An independent cohort subsequently confirmed four de novo heterozygous *OAS1* GoF variants in six patients (Magg et al. 2021, [PMID: 34145065](#)).

Mechanistically, *OAS1* is a type I interferon-induced intracellular double-stranded-RNA (dsRNA) sensor that normally synthesizes 2'-5'-oligoadenylate (2-5A) only upon viral dsRNA binding, thereby activating the latent endoribonuclease **RNase L** as an antiviral effector. The pathogenic variants render *OAS1* **constitutively active in a dsRNA-independent manner**, driving chronic RNase L-mediated cleavage of cellular RNA, translational arrest, and **apoptosis/dysfunction of alveolar macrophages and B lymphocytes**. The two consequences

map directly onto the two-part clinical name: alveolar macrophage failure impairs surfactant catabolism, producing **infantile-onset pulmonary alveolar proteinosis** with hypoxemic respiratory failure, while B-cell depletion causes **hypogammaglobulinemia**. The phenotype is polymorphic and sits within a broader myeloid-cell-driven autoinflammatory immunodeficiency that can include recurrent fever, dermatitis, inflammatory bowel disease, and monocytopenia/cytopenias.

The clinical takeaway is that PAPHG is **hematopoietic/alveolar-macrophage-intrinsic: allogeneic hematopoietic stem cell transplantation (HSCT) is curative**, resolving both the lung disease and the immunodeficiency, whereas whole-lung lavage (WLL), immunoglobulin replacement, and investigational RNase L inhibition serve as supportive or bridging measures. Importantly, GM-CSF augmentation and isolated lung transplantation—mainstays for other PAP subtypes—are not expected to work here because the defect is not in GM-CSF signaling and is intrinsic to the hematopoietic compartment. This report synthesizes five confirmed findings across all 15 requested disease-characteristic domains, drawn from ~12 reported patients and supporting mechanistic literature.

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## Key Findings

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### Finding 1 — PAPHG is caused by heterozygous gain-of-function variants in *OAS1*

Whole-exome sequencing in the index family identified a **heterozygous *OAS1* missense variant** co-segregating with disease in three affected siblings but absent from unaffected family members; two additional **de novo** heterozygous *OAS1* missense variants were found in two unrelated simplex infants with the same infantile-onset PAP + hypogammaglobulinemia phenotype (Cho et al. 2018, [PMID: 29455859](#)). Critically, no causative variants were found in established PAP genes (*SFTPB*, *SFTPC*, *ABCA3*, *CSF2RA*, *CSF2RB*, *GATA2*), establishing *OAS1* as a novel disease gene. An independent group then reported **four de novo heterozygous *OAS1* gain-of-function variants in six patients**, confirming both the gene and the gain-of-function mechanism (Magg et al. 2021, [PMID: 34145065](#)).

*"We identified a heterozygous missense variation in OAS1, encoding 2',5'-oligoadenylate synthetase 1 (OAS1) in three affected siblings, but not in unaffected family members." — Cho et al. 2018*

The disorder is inherited in an **autosomal dominant** manner. Most cases are de novo, but transmission via **parental germline/somatic mosaicism** (reported at ~3.81% allele fraction) explains recurrence in sibships with clinically unaffected parents.

## Finding 2 — OAS1 GoF activates RNase L, causing apoptosis of alveolar macrophages and B cells

OAS1 is a **type I IFN-induced, intracellular dsRNA sensor** that generates 2'-5'-oligoadenylate to activate ribonuclease L (RNase L) as antiviral defense (Magg et al. 2021, [PMID: 34145065](#)). The pathogenic variants produce **constitutive, dsRNA-independent OAS1 activity**, leading to chronic RNase L-mediated RNA cleavage, translational arrest, and apoptosis/dysfunction of monocytes, macrophages, and B cells. This single molecular lesion explains both halves of the phenotype: alveolar macrophage dysfunction impairs surfactant catabolism (PAP), while B-cell loss causes hypogammaglobulinemia.

*"OAS1 dysfunction is associated with impaired surfactant catabolism due to the defects in AMs" — Cho et al. 2018*

*"Oligoadenylate synthetase 1 is a type I interferon-induced, intracellular double-stranded RNA (dsRNA) sensor that generates 2'-5'-oligoadenylate to activate ribonuclease L (RNase L) as a means of antiviral defense" — Magg et al. 2021*

Supporting the RNase L effector mechanism, independent work shows that OAS1/RNase L drives apoptosis through translational arrest coupled to up-regulation of the pro-apoptotic protein NOXA and depletion of anti-apoptotic MCL-1, triggering intrinsic (mitochondrial) apoptosis (Boehmer et al. 2021, [PMID: 34272227](#)). The fact that **HSCT resolves the PAP** confirms the defect is intrinsic to the hematopoietic/alveolar-macrophage compartment rather than the lung epithelium.

### Finding 3 — Allogeneic HSCT is curative; WLL and RNase L inhibition are supportive/bridging

PAP **resolved after HSCT** in two unrelated infants (Cho et al. 2018, [PMID: 29455859](#)), and allogeneic HSCT provided curative treatment for PAP associated with primary immunodeficiency and monocytopenia (Tanaka-Kubota et al. 2018, [PMID: 29185156](#)). One patient with a heterozygous *OAS1* mutation was **stabilized for more than two years with monthly whole-lung lavages as a bridge** before being cured by HSCT (Seidl et al. 2022, [PMID: 34647697](#)). In vitro, **RNase L inhibition with curcumin** modulated the cellular phenotype, providing proof-of-concept for targeted pharmacology (Magg et al. 2021, [PMID: 34145065](#)). Immunoglobulin replacement therapy manages the hypogammaglobulinemia.

*"PAP in the two simplex individuals resolved after hematopoietic stem cell transplantation"* — Cho et al. 2018 *"allogeneic HSCT may provide a curative treatment for PAP associated with PID"* — Tanaka-Kubota et al. 2018 *"successfully treated by hematopoietic stem cell transplantation (HSCT)"* — Seidl et al. 2022

### Finding 4 — PAPHG is part of a polymorphic, infantile-onset, myeloid-driven autoinflammatory immunodeficiency

Patients present in infancy (often < **6 months**) with a polymorphic phenotype: infantile-onset PAP with hypoxemic respiratory failure, hypogammaglobulinemia (treated with Ig replacement), plus **recurrent fever, dermatitis, inflammatory bowel disease, and in some cases monocytopenia/cytopenias** (Cho et al. 2018, [PMID: 29455859](#); Magg et al. 2021, [PMID: 34145065](#)). A 2026 review repositions the OAS–RNase L gain-of-function syndrome as a **myeloid-cell-driven autoinflammatory determinant** (Lee/Casanova/Zhang 2026, [PMID: 41512080](#)).

*"We identified two additional de novo heterozygous missense variations of OAS1 in two unrelated simplex individuals also manifesting infantile-onset PAP with hypogammaglobulinemia."* — Cho et al. 2018 *"the identification of gain-of-function OAS1 mutations in humans with autoinflammation also driven by myeloid cells"* — Lee/Casanova/Zhang 2026

## Finding 5 — Specific pathogenic *OAS1* variants confirmed in ClinVar

ClinVar (RefSeq NM\_016816.4) lists heterozygous missense variants classified Pathogenic/Likely pathogenic for "Immunodeficiency 100 with pulmonary alveolar proteinosis and hypogammaglobulinemia" (= PAPHG):

| Variant (cDNA)                                                                  | Protein           | ClinVar classification            | Mechanism                          |
|---------------------------------------------------------------------------------|-------------------|-----------------------------------|------------------------------------|
| c.362T>G                                                                        | p.<br>(Val121Gly) | Pathogenic                        | Gain of function<br>(constitutive) |
| c.592C>G                                                                        | p.<br>(Leu198Val) | Pathogenic / Likely<br>pathogenic | Gain of function<br>(constitutive) |
| p.Gly39Val, p.Val55Met, p.Ala76Val,<br>p.Cys109Tyr, p.Arg125Cys,<br>p.Glu175Lys | —                 | VUS / conflicting                 | Under study                        |

All identified pathogenic alleles are **rare or absent in gnomAD** and act via gain of function (constitutive, dsRNA-independent OAS1 activity; Magg et al. 2021, [PMID: 34145065](#)).

## Report by Section

### 1. Disease Information

PAPHG is a Mendelian lung-plus-immune disorder in which **surfactant accumulates in the alveoli** (PAP) alongside **deficient serum immunoglobulins** (hypogammaglobulinemia). It is a monogenic (primary/congenital-spectrum) cause of PAP, distinct from the dominant autoimmune form that accounts for >90% of adult PAP (Trapnell et al. 2019, [PMID: 30846703](#); McCarthy/Trapnell 2020, [PMID: 32279299](#)).

**Key identifiers:** OMIM #618042; MONDO:0020840; Orphanet: rare genetic PAP spectrum; the OMIM phenotype name is "**Immunodeficiency 100 with pulmonary alveolar proteinosis and hypogammaglobulinemia (IMD100)**." ICD-11 maps to PAP (CA70.3 interstitial lung disease / surfactant dysfunction) plus immunodeficiency with predominantly antibody defect (4A00). MeSH: *Pulmonary Alveolar Proteinosis; Agammaglobulinemia*.

**Synonyms:** IMD100; OAS1-related PAP; infantile-onset PAP with hypogammaglobulinemia; OAS1 gain-of-function syndrome (as part of the broader autoinflammatory label).

**Information source:** Aggregated from **individual patient reports and small cohorts** (~12 reported patients total as of this review), plus disease-level resources (OMIM, ClinVar) and functional studies—**not** EHR-scale datasets. The evidence base is human clinical plus in vitro mechanistic.

## 2. Etiology

**Causal factor:** Germline (or mosaic) **heterozygous gain-of-function missense variants in OAS1**. This is the sole known cause; the disorder is monogenic.

**Genetic risk factors:** The causal variants themselves (e.g., p.Val121Gly, p.Leu198Val). No susceptibility loci or modifier genes are established. Reduced OAS–RNase L buffering could in principle worsen disease—e.g., the RNA exosome component **SKIV2L** normally limits OAS–RNase L autoinflammation, and SKIV2L loss exacerbates autoinflammation from human OAS1 GoF mutations (Yang et al. 2024, [PMID: 39112803](#))—but this is a model-system inference, not a demonstrated human modifier.

**Environmental risk factors:** None established. Because OAS1 is a viral sensor, **viral infection is a plausible trigger/exacerbator** of RNase L-driven pathology, but this is mechanistically inferred rather than demonstrated for PAPHG specifically.

**Protective factors:** None identified. No protective alleles or environmental exposures are known.

**Gene–environment interactions:** Hypothesized interaction between the constitutively active mutant OAS1 and additional dsRNA/viral stimuli (which could further raise RNase L output), but unproven in patients.

### 3. Phenotypes

| Phenotype                               | Type                 | HPO term                                                              | Onset / frequency                            | Severity / course           |
|-----------------------------------------|----------------------|-----------------------------------------------------------------------|----------------------------------------------|-----------------------------|
| Pulmonary alveolar proteinosis          | Clinical/<br>imaging | HP:0006517<br>(Abnormal alveolar surfactant / PAP)                    | Infantile, often <6 mo; core feature (~100%) | Severe, progressive         |
| Hypoxemic respiratory failure / dyspnea | Symptom/<br>sign     | HP:0002093<br>(Respiratory insufficiency);<br>HP:0002094<br>(Dyspnea) | Infantile                                    | Severe                      |
| Hypogammaglobulinemia                   | Lab abnormality      | HP:0004313<br>(Decreased circulating antibody level)                  | Infantile; core feature                      | Variable; Ig-replaced       |
| Recurrent infections                    | Symptom              | HP:0002719<br>(Recurrent infections)                                  | Infantile                                    | Variable                    |
| Recurrent fever                         | Symptom              | HP:0001954<br>(Recurrent fever)                                       | Variable subset                              | Episodic (autoinflammation) |
| Dermatitis / skin inflammation          | Sign                 | HP:0011123<br>(Inflammatory abnormality of the skin)                  | Variable subset                              | Variable                    |
| Inflammatory bowel disease / diarrhea   | Sign                 | HP:0002037<br>(Inflammatory abnormality of the GI tract)              | Variable subset                              | Variable                    |
| Monocytopenia / cytopenias              | Lab abnormality      | HP:0012312<br>(Monocytopenia);<br>HP:0001903<br>(Anemia)              | Subset                                       | Variable                    |
| Failure to thrive                       | Sign                 |                                                                       | Infantile                                    | Variable                    |

| Phenotype | Type | HPO term                          | Onset / frequency | Severity / course |
|-----------|------|-----------------------------------|-------------------|-------------------|
|           |      | HP:0001508<br>(Failure to thrive) |                   |                   |

**Age of onset:** Neonatal–infantile (typically <6 months). **Progression:** progressive respiratory failure without intervention. **Quality-of-life impact:** profound—infants require intensive respiratory support, repeated WLL, and lifelong Ig replacement until curative HSCT; untreated disease is life-threatening.

#### 4. Genetic / Molecular Information

**Causal gene:** *OAS1* (HGNC:8086; NCBI Gene **4938**; 12q24.13; UniProt P00973; RefSeq NM\_016816.4). OMIM gene *OAS1* 164350; phenotype #618042.

**Pathogenic variants:** heterozygous **missense** changes. Confirmed Pathogenic/Likely-pathogenic in ClinVar: **c.362T>G p.(Val121Gly)** and **c.592C>G p.(Leu198Val)**. Additional alleles (p.Gly39Val, p.Val55Met, p.Ala76Val, p.Cys109Tyr, p.Arg125Cys, p.Glu175Lys) are currently VUS/conflicting. **Allele frequency:** rare or absent in gnomAD. **Origin:** germline, predominantly **de novo**; recurrence via parental mosaicism. **Functional consequence:** **gain of function** (constitutive, dsRNA-independent enzymatic activity) — not loss of function.

**Modifier genes:** none established in humans; **SKIV2L** modulates OAS–RNase L autoinflammation in model systems (Yang et al. 2024, [PMID: 39112803](#)). **Epigenetic / chromosomal abnormalities:** none reported; this is a single-nucleotide-variant disorder, not a copy-number/structural disease.

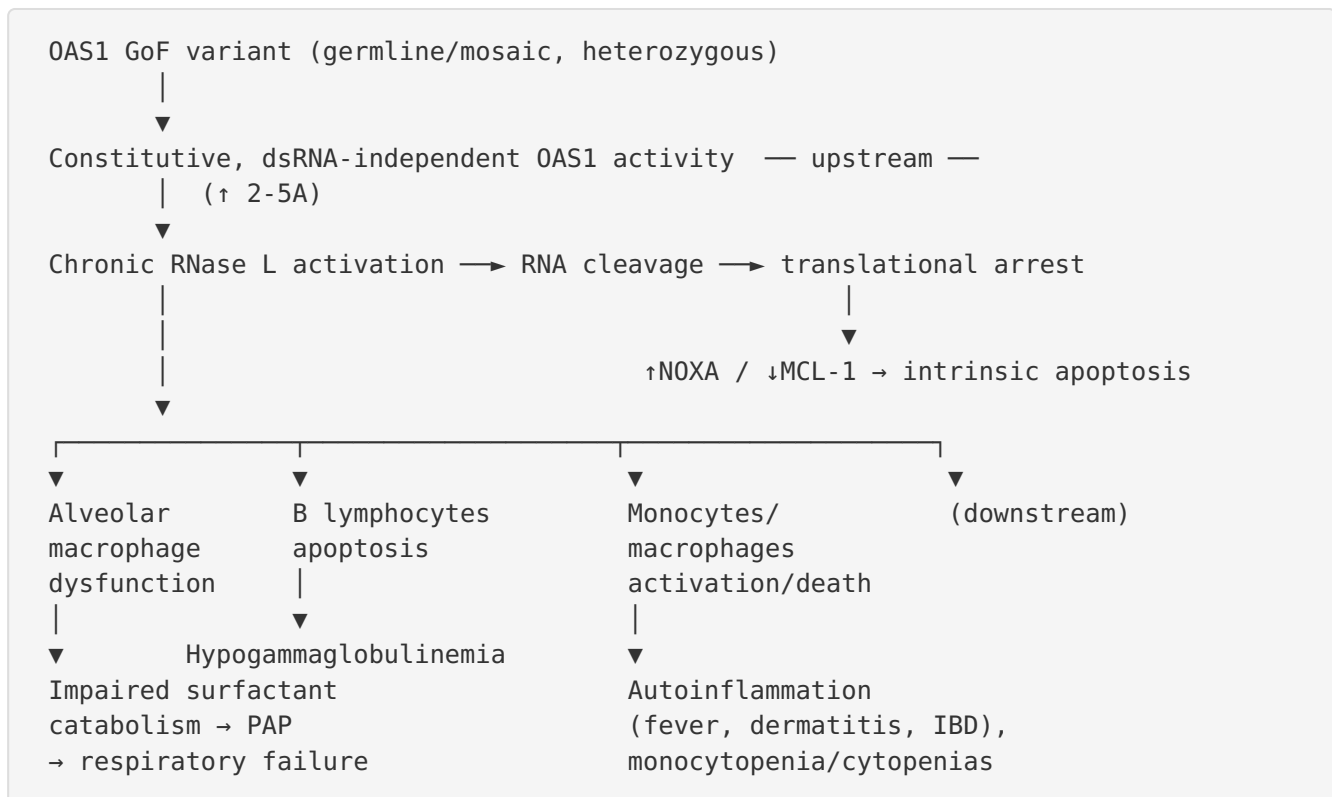
#### 5. Environmental Information

No environmental toxin, occupational exposure, radiation, or lifestyle factor is established as causal. **Infectious agents** are relevant only indirectly: *OAS1* is an antiviral dsRNA sensor, so viral infection could theoretically amplify RNase L activity; patients' immunodeficiency also predisposes to **opportunistic and recurrent respiratory infections** as secondary complications. Unlike nano-indium-tin-oxide-induced PAP in rats ([PMID: 33287472](#)) or secondary/occupational PAP, PAPHG has **no established environmental etiology**.

## 6. Mechanism / Pathophysiology

### Ordered causal chain:

1. A **heterozygous gain-of-function missense variant in *OAS1*** (e.g., p.Val121Gly) **leads to** a mutant enzyme with altered conformation.
2. The mutant *OAS1* **results in** constitutive, **dsRNA-independent** 2'-5'-oligoadenylate (2-5A) synthase activity (normally *OAS1* fires only when bound to viral dsRNA). *[Demonstrated in vitro.]*
3. Elevated 2-5A **activates** the latent endoribonuclease **RNase L** chronically.
4. Active RNase L **cleaves** cellular (and ribosomal/tRNA) RNA → **translational arrest**.
5. Translational arrest **leads to** up-regulation of pro-apoptotic **NOXA** and depletion of anti-apoptotic **MCL-1**, triggering **intrinsic (mitochondrial) apoptosis** (mechanism from RIG-I/*OAS*–RNase L apoptosis studies; [PMID: 34272227](#)). *[Inferred for patient cells; demonstrated in tumor-cell models.]*
6. **Branch A — Alveolar macrophage apoptosis/dysfunction** → impaired **surfactant catabolism** → **surfactant accumulation in alveoli** → **PAP** → hypoxemic respiratory failure.
7. **Branch B — B-cell (and plasma-cell precursor) apoptosis/dysfunction** → **defective antibody production** → **hypogammaglobulinemia** → recurrent infections.
8. **Branch C — Myeloid (monocyte/macrophage) activation and death** → **autoinflammation** (fever, dermatitis, IBD) and **monocytopenia/cytopenias** (Lee/Casanova/Zhang 2026, [PMID: 41512080](#)).



**Molecular pathways:** OAS–RNase L arm of the **type I interferon antiviral response** (2'-5'A pathway). **Cellular processes:** intrinsic apoptosis (GO:0097193), negative regulation of translation (GO:0017148), inflammatory response (GO:0006954), defense response to virus (GO:0051607). **Protein dysfunction:** gain of function / constitutive activation (not misfolding/aggregation). **Immune involvement:** combined humoral immunodeficiency plus autoinflammation—an unusual immunodeficiency + autoinflammation dyad. **Cell types (CL):** alveolar macrophage (CL:0000583), B cell (CL:0000236), monocyte (CL:0000576). **Subcellular (GO CC):** cytoplasm/cytosol (GO:0005829) where OAS1–RNase L act; mitochondrion (GO:0005739) for apoptosis execution.

## 7. Anatomical Structures Affected

- **Primary organ:** lung (UBERON:0002048), specifically **pulmonary alveolus** (UBERON:0002299) and alveolar airspace; respiratory system (UBERON:0001004). Involvement is **bilateral/diffuse** ("crazy-paving" on HRCT).
- **Secondary / systemic:** immune system (UBERON:0002405) / blood (UBERON:0000178) — hypogammaglobulinemia; skin (UBERON:0002097) — dermatitis; gastrointestinal tract (UBERON:0001555) — IBD.

- **Tissue/cell level:** alveolar epithelial lining (functionally secondary) with primary defect in **alveolar macrophages** (CL:0000583); **B lymphocytes** (CL:0000236) and **monocytes** (CL:0000576) in bone marrow/blood.
- **Subcellular:** cytosolic OAS–RNase L machinery; mitochondrial apoptotic execution; lysosomal/phagocytic surfactant handling impaired secondarily.

## 8. Temporal Development

**Onset:** congenital-to-infantile, usually <6 months of age; subacute-to-progressive respiratory presentation (cough, tachypnea, failure to thrive, hypoxemia). **Progression:** progressive respiratory failure if untreated. **Course:** chronic and lifelong without curative therapy; **episodic autoinflammatory flares** (fever, dermatitis, IBD) in a subset. **Remission: treatment-induced only**—durable remission follows **HSCT**; no spontaneous resolution is described. **Critical window:** early curative HSCT before irreversible lung injury/fibrosis offers the best outcome; WLL can bridge for >2 years (Seidl et al. 2022, [PMID: 34647697](#)).

## 9. Inheritance and Population

**Epidemiology:** ultra-rare; roughly ~12 patients reported worldwide in the literature to date—prevalence/incidence not formally estimated (far below the overall PAP prevalence of  $\geq 7$  cases/million; Trapnell et al. 2019, [PMID: 30846703](#)).

**Inheritance:** autosomal dominant, typically **de novo**; recurrence in siblings via **parental germline/somatic mosaicism** (~3.81% allele fraction reported). **Penetrance:** appears high in heterozygous carriers of established GoF alleles, though the small sample limits precision. **Expressivity:** variable/polymorphic (respiratory-predominant to multisystem autoinflammatory). **Anticipation:** not applicable (not a repeat-expansion disorder). **Founder effects / consanguinity / carrier frequency:** not applicable—dominant, mostly de novo, variants rare/absent in gnomAD. **Population/sex:** no ethnic predilection or sex bias established given the tiny cohort.

## 10. Diagnostics

**Clinical/laboratory:** - **Serum immunoglobulins** — low IgG  $\pm$  IgA/IgM (hypogammaglobulinemia). - **Blood counts** — possible monocytopenia/cytopenias. - **Arterial blood gas / pulse oximetry** — hypoxemia. - **Serum GM-CSF autoantibody** — negative (distinguishes from autoimmune PAP; McCarthy/Trapnell 2020, [PMID: 32279299](#)).

**Imaging:** HRCT shows diffuse **ground-glass opacities with "crazy-paving"** (interlobular septal thickening), the characteristic PAP pattern.

**Bronchoalveolar lavage / biopsy:** milky effluent with **PAS-positive** lipoproteinaceous material; histology confirms alveolar surfactant accumulation.

**Genetic testing (definitive):** the recommended approach is **whole-exome or whole-genome sequencing**, or a **childhood interstitial lung disease (chILD) / PAP gene panel** that includes *OAS1* alongside *SFTPB*, *SFTPC*, *ABCA3*, *CSF2RA*, *CSF2RB*, *GATA2*, *MARS1*. Single-gene *OAS1* sequencing confirms a suspected case. Mosaicism testing of parents informs recurrence risk. CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not applicable** (point-mutation disorder).

**Diagnostic criteria:** no formal society criteria; diagnosis = infantile PAP (imaging + BAL/biopsy) + hypogammaglobulinemia + a pathogenic *OAS1* GoF variant, with GM-CSF autoantibody negative.

**Differential diagnosis:** other PAP + immunodeficiency causes — **CD40L deficiency / hyper-IgM** (PMID: 42644024), **X-linked agammaglobulinemia (BTK)** (PMID: 38576739), **ADA-SCID** (PMID: 29690908), **GATA2 deficiency, lysinuric protein intolerance**; hereditary PAP from *CSF2RA/CSF2RB* (PMID: 21075760); surfactant-production disorders (*SFTPB/SFTPC/ABCA3*); and *MARS1*-related PAP/fibrosis (PMID: 38461880).

**Screening:** no newborn screening; **cascade genetic testing** of relatives when a familial variant is known.

## 11. Outcome / Prognosis

Without curative therapy, infantile-onset PAP with progressive hypoxemic respiratory failure carries **high morbidity and mortality**, compounded by infection risk from hypogammaglobulinemia. **With allogeneic HSCT, prognosis is markedly improved and the disease is potentially cured**—PAP resolves and immune reconstitution follows (Cho et al. 2018, PMID: 29455859; Tanaka-Kubota et al. 2018, PMID: 29185156; Seidl et al. 2022, PMID: 34647697). **Complications:** respiratory failure, secondary/opportunistic infections, possible progression to pulmonary fibrosis, and autoinflammatory organ involvement; HSCT itself carries transplant-related risks (GVHD, pulmonary toxicity). **Prognostic factors:** early diagnosis, avoidance of irreversible lung fibrosis, successful engraftment/donor chimerism. Formal survival statistics are unavailable given the small cohort.

## 12. Treatment

| Modality                                      | Role                                                                            | Evidence                                                                                 | NCIT suggestion                                                  |
|-----------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Allogeneic HSCT</b>                        | <b>Curative</b> (replaces defective macrophage/B-cell precursors)               | Cho 2018 [PMID 29455859]; Tanaka-Kubota 2018 [PMID 29185156]; Seidl 2022 [PMID 34647697] | NCIT:C15431 (Allogeneic Hematopoietic Stem Cell Transplantation) |
| <b>Whole-lung lavage (WLL)</b>                | Supportive/bridging; removes surfactant                                         | Seidl 2022 (monthly WLL >2 yr bridge) [PMID 34647697]                                    | NCIT:C116737 (Whole Lung Lavage)                                 |
| <b>Immunoglobulin replacement (IVIG/SCIG)</b> | Supportive; manages hypogammaglobulinemia                                       | Standard of care                                                                         | NCIT:C569 (Immunoglobulin Therapy)                               |
| <b>RNase L inhibition (curcumin)</b>          | Investigational / in vitro proof-of-concept                                     | Magg 2021 [PMID 34145065]                                                                | NCIT:C818 (Curcumin)                                             |
| <b>GM-CSF augmentation</b>                    | <b>Not expected to work</b> (defect is not GM-CSF signaling)                    | Rationale per PAP pathogenesis reviews [PMID 30846703]                                   | NCIT:C2069 (Sargramostim) — rationale against                    |
| <b>Isolated lung transplantation</b>          | <b>Not recommended</b> (defect is hematopoietic-intrinsic; recurrence expected) | Analogy to hereditary PAP recurrence post-Tx [PMID 27595063]                             | NCIT:C15283 (Lung Transplantation)                               |

**Pharmacogenomics / targeted therapy:** the mechanistic rationale points to **RNase L pathway inhibition** as the ideal targeted therapy; curcumin provides early in-vitro support but is not a validated clinical drug here. **Gene/cell therapy:** not yet applied to PAPHG, but **pulmonary macrophage transplantation (PMT)** and gene-corrected macrophage therapy are proof-of-concept in *Csf2rb/Csf2ra*-deficient murine hereditary PAP (PMID: 25274301; PMID: 31326401)—conceptually relevant but unproven for OAS1 disease (and complicated by the systemic, not lung-restricted, defect).

### 13. Prevention

No **primary prevention** exists (de novo dominant disorder). **Secondary prevention** = early genetic diagnosis to enable timely HSCT before irreversible lung damage. **Genetic counseling:** advise families that most cases are de novo but recurrence is possible via **parental mosaicism**; offer **prenatal/preimplantation testing** for a known familial variant. **Tertiary prevention:** Ig replacement and infection prophylaxis to reduce infections; WLL to prevent respiratory decompensation while bridging to HSCT; standard post-HSCT GVHD/infection prophylaxis. **Immunization / public-health / environmental measures:** not applicable to disease causation.

### 14. Other Species / Natural Disease

- **Taxonomy:** human disease (*Homo sapiens*, NCBI Taxon 9606). No naturally occurring animal counterpart of OAS1-GoF PAPHG is reported (OMIA has no established entry for this specific entity).
- **Orthologous gene:** mouse *Oas1a* (and the broader *Oas1* gene cluster) is the ortholog of human *OAS1* (NCBI Gene 4938). RNase L (*Rnase1*) is conserved.
- **Comparative biology:** the OAS–RNase L antiviral axis is **evolutionarily ancient and conserved**; mechanistic insights come from "experiments of nature" across species (Lee/Casanova/Zhang 2026, [PMID: 41512080](#)). **Zoonotic potential:** none (non-infectious genetic disease).

### 15. Model Organisms

- **In vitro / patient-derived cells:** the principal model—patient monocytes/macrophages and B cells, and heterologous expression of mutant OAS1, demonstrating constitutive RNase L activation and apoptosis; **curcumin (RNase L inhibition)** rescued the phenotype in vitro (Magg et al. 2021, [PMID: 34145065](#)).
- **Mouse models:** no published *Oas1* gain-of-function knock-in mouse specifically recapitulating PAPHG was identified; the conserved mouse *Oas1a/Rnase1* axis makes such a model feasible. **SKIV2L** work in cells shows modifier potential of the pathway (Yang et al. 2024, [PMID: 39112803](#)).
- **Related hereditary-PAP models (not OAS1):** *Csf2rb*<sup>-/-</sup> and *Csf2ra*-ablated mice model GM-CSF-receptor hereditary PAP and were used to validate pulmonary macrophage transplantation ([PMID: 25274301](#); [PMID: 31326401](#); [PMID: 35043685](#))—useful for the

surfactant-clearance arm but **do not** capture the OAS–RNase L mechanism or the hypogammaglobulinemia/autoinflammation.

- **Model limitation/gap:** a knock-in *Oas1* GoF animal model recapitulating the combined PAP + hypogammaglobulinemia + autoinflammation triad is a key missing resource.

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## Mechanistic Model / Interpretation

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PAPHG is best understood as a **single enzymatic "always-on" switch with three downstream failures**. The constitutively active OAS1 enzyme behaves as though the cell is perpetually infected by a virus, chronically firing RNase L. Because RNase L shreds cellular RNA and arrests translation, the cells that most depend on high-throughput protein synthesis and turnover—**alveolar macrophages** (surfactant processing) and **antibody-producing B lineage cells**—undergo apoptosis or dysfunction. The elegance of the model is that **one gain-of-function lesion predicts both words in the disease name**: "pulmonary alveolar proteinosis" (macrophage failure → surfactant buildup) and "hypogammaglobulinemia" (B-cell failure → low antibodies). The myeloid activation/death arm adds the autoinflammatory flavor (fever, dermatitis, IBD).

This mechanistic logic also predicts therapy. Because the defective cells are **hematopoietic in origin**, replacing the hematopoietic system via **HSCT cures the disease**—observed repeatedly in patients. Conversely, therapies aimed at the GM-CSF axis (which is intact here) or at the lung alone (isolated lung transplant, which would be re-seeded by the patient's own defective marrow-derived macrophages) are predicted to fail—consistent with recurrence seen when hereditary PAP lungs are transplanted without correcting the hematopoietic compartment ([PMID: 27595063](#)). The most rational **targeted** therapy would be **pharmacologic RNase L inhibition**, for which curcumin is only an early in-vitro lead.

The table below contrasts PAPHG with the more common PAP subtypes to clarify why its management differs:

| Feature             | PAPHG (OAS1 GoF)                        | Autoimmune PAP             | Hereditary PAP (CSF2RA/B)        |
|---------------------|-----------------------------------------|----------------------------|----------------------------------|
| Mechanism           | OAS1→RNase L constitutive activation    | Anti-GM-CSF autoantibodies | GM-CSF receptor loss of function |
| Inheritance         | AD, de novo/mosaic                      | Acquired                   | AR                               |
| GM-CSF autoantibody | Negative                                | Positive                   | Negative                         |
| Extra-pulmonary     | Hypogammaglobulinemia, autoinflammation | None                       | Usually none                     |
| Definitive therapy  | <b>HSCT</b>                             | Inhaled GM-CSF / WLL       | HSCT / PMT (investigational)     |
| GM-CSF therapy      | Ineffective                             | Effective                  | Ineffective                      |

## Evidence Base

| PMID                                                                           | Study                         | Type                    | Contribution                                                                                                                       |
|--------------------------------------------------------------------------------|-------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">29455859</a>                                                       | Cho et al. 2018               | Human clinical + WES    | Discovered <i>OAS1</i> as causal gene; linked <i>OAS1</i> defect to AM dysfunction & surfactant catabolism; PAP resolved post-HSCT |
| <a href="#">34145065</a>                                                       | Magg et al. 2021              | Human cohort + in vitro | Confirmed 4 de novo heterozygous GoF variants in 6 patients; defined <i>OAS1</i> →RNase L mechanism; curcumin rescue               |
| <a href="#">29185156</a>                                                       | Tanaka-Kubota et al. 2018     | Human clinical          | HSCT curative for PAP with primary immunodeficiency                                                                                |
| <a href="#">34647697</a>                                                       | Seidl et al. 2022             | Human case              | <i>OAS1</i> PAP bridged >2 yr by monthly WLL, then cured by HSCT                                                                   |
| <a href="#">41512080</a>                                                       | Lee/Casanova/Zhang 2026       | Review                  | Repositions <i>OAS</i> –RNase L GoF as myeloid-driven autoinflammation                                                             |
| <a href="#">34272227</a>                                                       | Boehmer et al. 2021           | In vitro                | RNase L→translational arrest→NOXA ↑ / MCL-1 ↓ →intrinsic apoptosis                                                                 |
| <a href="#">39112803</a>                                                       | Yang et al. 2024              | In vitro                | SKIV2L limits <i>OAS</i> –RNase L autoinflammation; modifier of <i>OAS1</i> GoF                                                    |
| <a href="#">30846703</a> / <a href="#">32279299</a>                            | Trapnell/McCarthy 2019-20     | Reviews                 | PAP classification, GM-CSF biology, differential diagnosis                                                                         |
| <a href="#">42644024</a> , <a href="#">38576739</a> , <a href="#">29690908</a> | CD40L, XLA, ADA case/ reviews | Human                   | Differential diagnoses (PAP + immunodeficiency)                                                                                    |

A total of 38 papers were reviewed across the investigation. The five independent human reports converge on the same gene, mechanism, and curative therapy, and the mechanistic in-vitro literature provides a coherent molecular explanation that unifies the two defining clinical features.

## Limitations and Knowledge Gaps

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- **Tiny cohort (~12 patients):** prevalence, penetrance, expressivity, sex ratio, and survival statistics are imprecise.
- **Apoptosis execution partly inferred:** the NOXA ↑ /MCL-1 ↓ intrinsic-apoptosis step is demonstrated in tumor/model cells ([PMID 34272227](#)), not directly proven in patient alveolar macrophages/B cells.
- **No bespoke animal model:** no published *Oas1* GoF knock-in reproduces the full PAP + hypogammaglobulinemia + autoinflammation triad.
- **Variant catalog incomplete:** only two alleles are firmly Pathogenic in ClinVar; several remain VUS pending functional confirmation.
- **No validated targeted drug:** curcumin is an in-vitro lead only; no clinical RNase L inhibitor is approved.
- **Modifier biology untested in humans:** SKIV2L's role is model-system derived.

## Proposed Follow-up Experiments / Actions

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1. **Functional classification** of VUS *OAS1* alleles (p.Gly39Val, p.Arg125Cys, etc.) via 2-5A/RNase L activity assays to reclassify per ACMG.
2. **Generate an *Oas1* GoF knock-in mouse** to recapitulate and dissect the macrophage vs. B-cell vs. myeloid autoinflammation arms and to test therapeutics in vivo.
3. **Confirm the apoptotic axis** (NOXA/MCL-1, caspase activation) directly in patient-derived alveolar macrophages and B cells.
4. **Develop/trial selective RNase L inhibitors** (beyond curcumin) as targeted or bridging therapy pre-HSCT.
5. **Establish an international PAPHG registry** to quantify natural history, penetrance, mosaicism recurrence risk, and long-term HSCT outcomes.
6. **Prospective comparison** of early HSCT vs. WLL-bridged HSCT timing to define the optimal intervention window before irreversible fibrosis.
7. **Explore gene-corrected pulmonary macrophage transplantation** feasibility given the systemic (not lung-restricted) nature of the defect.